

For temporal grounding tasks, we found that enhancing task diversity has a significant effect on improving the model’s temporal perception generalization ability. We can boldly speculate that if there are sufficiently diverse training data task types, most tasks with relatively simple output formats can achieve results comparable to expert models through appropriate instruction fine-tuning.

Through the integration of diverse expert tasks and the optimization of language representations, MLLMs can achieve substantial improvements in their overall capabilities. This allows them to effectively comprehend and address complex tasks, rivaling or even exceeding the performance of specialized expert models within specific domains. Looking ahead, MLLMs have the potential to evolve into highly versatile AI models, transcending traditional conversational and QA capabilities. They will be equipped to handle a wide range of complex expert tasks across various domains, such as vision, language, and reasoning.

G.2 WHY DOES TEMPORAL GROUNDING DATA LEAD TO ACCURACY LOSS IN SHORT-TERM VIDEOS?

We conducted ablation experiments using different combinations of temporal grounding data and regular data. The accuracy of VideoChat-T on MVBench after fine-tuning with various data combinations is shown in Table 16.

FT Data	MVBench (AVG)
TimeIT	54.7
TimeIT+Normal	55.3
Normal	56.1
TimePro	57.4
TimePro+Normal (Ours)	59.9

Table 16: Performance VideoChat-T on MVBench under different fine-tuning data settings.

The diversity of grounding data formats in the past has often been limited, which can lead to overfitting on Temporal Grounding tasks and cause the model to lose its general question-answering capability. We compared the TimeIT dataset proposed in TimeChat (Ren et al., 2024) with our TimePro dataset on MVBench. As shown in the Table 16, fine-tuning with only TimeIT resulted in the lowest accuracy, and the combined use of TimeIT+Normal also performed slightly worse than using Normal alone. This indicates that monotonous grounding data indeed damages the model’s original performance (as shown in Figure 1 at the beginning of the paper, TimeChat loses some of its general question-answering capability after fine-tuning, where it outputs localization times for general questions).

In contrast, our TimePro dataset includes diverse data, encompassing 9 different task types from 15 datasets, which helps mitigate the generalization loss caused by homogeneous grounding data types. Additionally, our dataset integrates Grounding with various general tasks. For instance, Grounded Caption requires detailed descriptions of corresponding video segments, while Reasoning Temporal Localization demands the model to reason about questions. This approach significantly enhances the model’s generalization ability and minimizes the impact on its original capability (e.g., short video accuracy). As demonstrated in the Table 16, the performance of using only TimePro exceeds that of using Normal alone, and the combined use of TimePro and Normal far surpasses all other combinations. This also confirms that our TimePro effectively preserves the model’s original performance.

Overall, using a single type of expert task training data can easily lead to model overfitting, resulting in significant loss of the model’s original capabilities. To preserve the model’s foundational generalization abilities, it is essential to use diversified training data. Additionally, incorporating data of various types and distributions, such as text, images, and videos, can further enhance the model’s generalization capabilities.

G.3 COULD TRAINING THE MODEL ON BOTH TEMPORAL AND NON-TEMPORAL GROUNDING DATA MITIGATE PERFORMANCE LOSS IN SHORT-TERM VIDEOS?

To address this question, we conducted additional ablation experiments. By training VideoChat-T with different combinations of temporal and non-temporal grounding data, we were able to clearly observe the effects of both types of data on the model’s performance. The results of the experiments are shown in the Table 17.

FT Data	MVBench	VideoMME	Charades-STA
	Avg	w/o subs	R1@0.5
Normal	56.1	42.6	8.0
TimePro	57.4	46.0	45.6
TimePro+Normal (Ours)	59.9	46.3	48.7

Table 17: Performance comparison of VideoChat-T using different combinations of temporal grounding and non-temporal grounding data.

It can be observed that the combined use of TimePro+Normal for VideoChat-T achieves the highest performance in short video QA, long video QA, and temporal grounding tasks. This not only demonstrates that using both temporal grounding and non-temporal grounding data can reduce performance loss in short videos, but also reveals that the effects of temporal and non-temporal grounding data are complementary across various tasks. The distinct differences between temporal grounding and non-temporal grounding tasks can respectively compensate for the model’s shortcomings in different task perspectives and feature distributions. The simultaneous use of both types of data can effectively enhance the model’s overall capabilities.

H CASE STUDY

H.1 MORE QUALITATIVE ANALYSIS

To further qualitatively analyze our model, we supplemented it with three types of examples. These examples are about long video QA, short video QA, and captioning tasks, all of which include temporal grounding.

More qualitative comparisons about long video QA are shown in Figure 6. VideoChat-T effectively handles various questions across different domains. By better perceiving the temporal relationships of different events occurring in long videos, it can more accurately and deeply understand the detailed content of the entire video.

More qualitative comparisons about short video QA are shown in Figure 7. VideoChat-T effectively retains the original capabilities of the base model. Through parameter-efficient initialization methods and appropriate training strategies, we minimize the damage to the base model’s capabilities caused by new architectures and data.

More qualitative comparisons about captioning are shown in Figure 8. Although VideoChat2 describes more local details in some scenarios compared to VideoChat-T, VideoChat-T focuses more on a series of temporal events, which aligns better with how humans typically describe videos.

H.2 SHORTCOMINGS

We also conducted a qualitative analysis of the shortcomings of VideoChat-T through examples. As shown in Figure 9, VideoChat-T performs poorly on examples with complex logic. In the left example, although VideoChat-T accurately identified the timing of the event, it failed to fully explain the motivation behind the man opening the isolation door, which was ”to fight the hijackers of the space elevator, seize the controller, and thus save the people in the entire space elevator.” In the right example, VideoChat-T correctly identified the event where Mr. Bean reached out to touch his desk mate’s table, but it incorrectly explained the true reason for this action, which was ”to cover up the fact that he was copying his desk mate’s exam by pretending to wipe dust off the desk.”

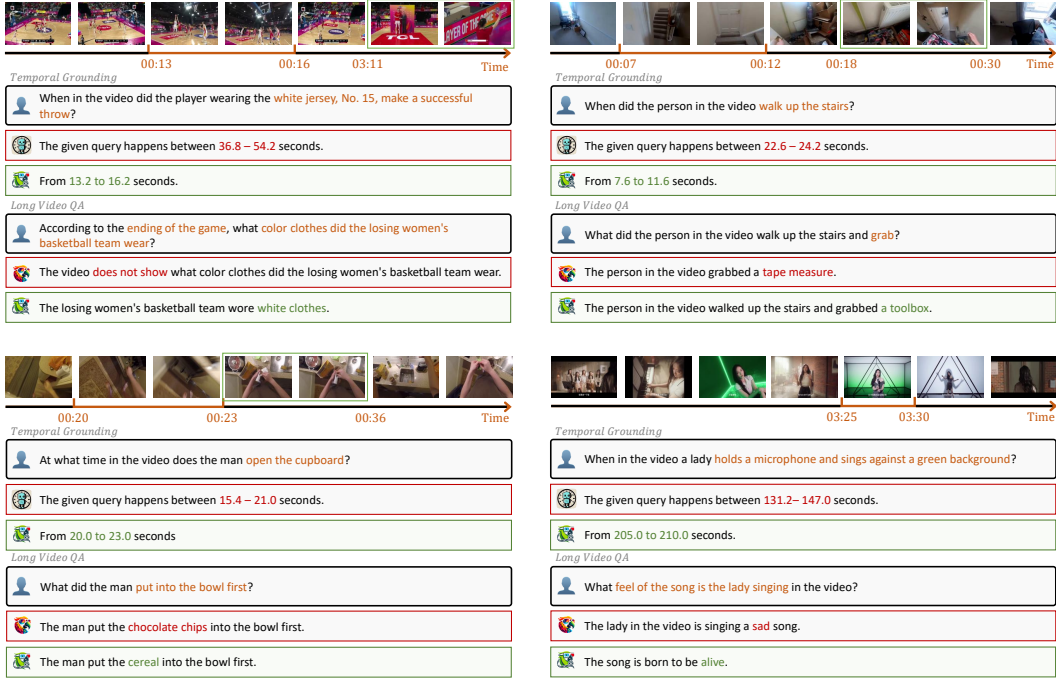


Figure 6: More qualitative comparisons in temporal grounding & long video QA.

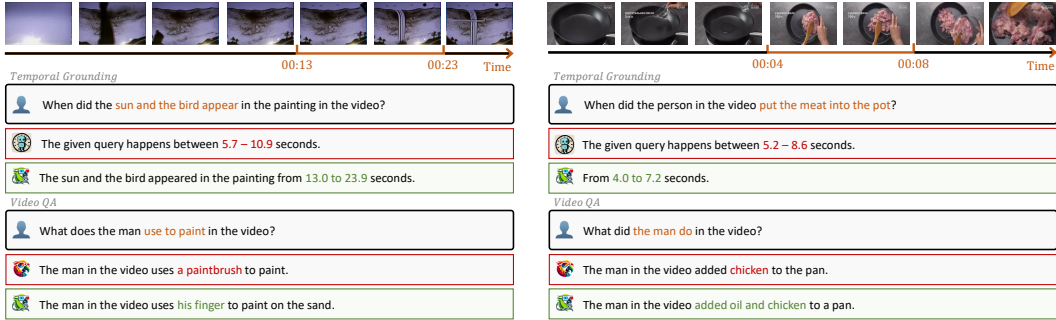


Figure 7: More qualitative comparisons in temporal grounding & short video QA.

Due to the preponderance of single-turn, perceptual questions in our training data and the lack of multi-step reasoning data with complex logic, our model struggles to handle more challenging scenarios that demand intricate logical reasoning. To address this limitation, we propose constructing data in a chain-of-thought format to guide the model through multi-step reasoning, enabling it to delve deeper into the underlying motivations and causal relationships within a video.